Stephen Bottos

Full-Stack Machine Learning Engineer · Low-Level Modeling to Production Systems & MLOps at Scale

778-966-8141 | bottos.steve@gmail.com | linkedin.com/in/stephen-bottos | github.com/stevebottos

Highlights

Full-stack ML engineer with 8+ years shipping production systems across vision, text, and audio. Equally at home in low-level modeling (custom architectures, quantization, distillation) and large-scale system and infrastructure design.

Patent holder (US11915434B2); built and own MLOps infrastructure supporting 1,000+ production models across 10k+ edge devices.

Building modern ML/AI systems end-to-end, spanning fine-tuning, distillation, LLMs, RAG, and multi-agent architectures, with recent work serving millions of requests daily.

Technical Skills

ML/AI: LLMs (RAG, prompt engineering, LoRA/QLoRA, prefix tuning), Knowledge Distillation, Quantization & Pruning, On-Device/Edge Inference, Entity Resolution, Multimodal Models (CLIP, BLIP, VideoLLava, custom implementations), Vision Models (CNNs, ViTs, VJepa2), Agentic Systems, Vector Databases
Frameworks: PyTorch, TensorFlow, Langchain, MLflow, Huggingface (Transformers, TEI), PySpark, Scikit-Learn, OpenCV, TensorRT, ONNX
Infrastructure: AWS (ECS, MWAA, ECR, Redshift, OpenSearch), GCP, Azure, Docker, Airflow, Git/GitHub CI/CD
Languages: Python, C/C++, SQL, Scala, MATLAB

Experience

Senior Machine Learning Engineer — Engine | Feb. 2026 – Present

Denver, CO (Remote)

- De-facto ML technical lead, directing engineers, data scientists, and annotators while owning the ML system end-to-end, from model architecture down to infrastructure specification. Define the GPU serving stack and CUDA/AMI requirements for Hugging Face TEI-based embedding deployment, which platform engineering provisions to spec.
- Lead ML engineer on a large-scale supplier room-mapping system (hotel inventory entity resolution) engineered to serve millions of requests daily in production, reconciling how hundreds of suppliers describe the same physical room into a single canonical entity. A core internal lever for rate optimization and catalog quality, and the basis of a forthcoming external mapping API revenue stream.
- Designed a knowledge-distillation architecture to make mapping tractable over a massive supplier corpus: an LLM teacher generates canonical-title targets from raw supplier strings, distilled into a small, low-latency student embedding model that collapses inference from per-query LLM generation to input string → embedding → nearest-neighbor lookup over an OpenSearch vector database.

Lead Machine Learning Engineer — Kibeam, Inc. | Aug. 2023 – Dec. 2025

Oakland, CA (Remote)

- Architected multimodal agentic systems integrating vision, text, and audio (Langchain, MLflow for GenAI, Huggingface, OpenAI, Gemini, ElevenLabs). For internal content-generation workflows, RAG-enabled reflection patterns spanning dozens of agents reduced content generation from days to under an hour.
- To overcome VLM spatial-reasoning limits, built a detection-driven grounding pipeline: an object detector localizes targets of interest in an unobstructed reference image, aligns them to an obstructed view, and supplies the VLM with focused crops to identify the obscured object, raising eval accuracy from 0.75 to 0.98.
- Built end-to-end MLOps infrastructure from scratch (AWS ECS, Airflow, MLflow, Redshift), which supports 100+ model training runs per week, and manages deployment and tracking of 1,000+ unique production models across 10k+ edge devices in the wild today. Contributed to Unity-based synthetic data generation as part of this system, achieving production-grade generalization across all models without manual real data collection.
- Designed the production CNN deployed across all ESP32 edge devices: adapted a ShuffleNetV2 backbone for int8 quantization and channels-last inference by inserting train-time channel-swap modules that fold into no-ops once quantized, removing channels-first/last permute overhead on device; also added GhostNet-style cheap feature generation early in the network, gaining several F1 points at negligible latency cost.
- Trained and deployed fine-tuned Vision Transformers and VLMs to analyze databases of internally collected images and propose soft-labels for regular re-training cycles as a form of offline knowledge distillation, ensuring continuous model improvement and eliminating manual annotation entirely.

- Led cross-team Machine Learning initiatives spanning 20+ employees, defining technical roadmaps, research direction, and development priorities across engineering and product teams. Directed advanced modeling and data curation efforts toward agentic automation to ensure direct alignment with key product vision and user needs.
- Supported the CEO directly in high-stakes technical demonstrations to audiences large and small, with full ownership of every ML/AI system component.

Senior Machine Learning Engineer — Plainsight, Inc. Aug. 2021 – Aug. 2023
San Diego, CA

- Designed, trained, and deployed custom PyTorch architectures driving business logic for customer accounts worth up to $10M; primarily Computer Vision, extending to video and text per use case.
- Pioneered use of foundation models (CLIP, Owl-ViT) for fine-tuning, zero-shot detection and auto-labeling. Leveraged early agentic systems like ViperGPT for compositional visual reasoning.
- Built and scaled cloud workflows in GCP, using custom Docker containers along with VertexAI to train and deploy models behind REST API endpoints for inference. Implemented CI/CD pipelines for automated model testing and deployment across multiple client environments.
- Extended internal data management software to support de-duplication, clustering, and exploration of millions of images/videos, accelerating data acquisition/QC workflows by 80% by leveraging embeddings and vector databases (Milvus, BigQuery, ClickHouse).
- Built Kubeflow and Apache Beam ETL pipelines handling large-scale data ingestion from client data lakes.
- Obtained Google Cloud Certified Professional Machine Learning Engineer certification.

Machine Learning Engineer — alwaysAI, Inc. June 2020 – Aug. 2021
San Diego, CA

- Designed and deployed production Computer Vision models (TensorFlow, PyTorch) with quantization and pruning for real-time edge inference. Built AWS ECS-based platform handling ETL pipelines, model training, and serving infrastructure.
- Sole inventor of a patented multi-camera object tracking and re-identification system (US11915434B2): a network of edge-deployed models tracking people across store cameras to measure interactions in regions of interest, turning physical spaces into digital-style metrics. Powered two contracts worth $1M+, still in use today.

Machine Learning Engineer — Qimia, Inc. Sep. 2019 – June 2020
San Diego, CA

- Drove the data engineering, data science, and ML work for a large advertising agency largely autonomously (originating most of the technical approach myself, with senior support rather than direction) to optimize regional campaigns across billions of data points on individuals nationwide. Also partnered with sub-contractors on early-stage spill-detection and inventory-analysis software running on robots deployed to grocery/big-box stores.
- Built PySpark/Scala pipelines and models on AWS Redshift for analytics, feature engineering, and predictive modeling at scale, surfacing results through custom dashboards and Dockerized applications.

Machine Learning Researcher — University of Windsor Jan. 2018 – Aug. 2019
Windsor, ON

- First-authored three publications on eye-gaze tracking using Hidden Markov Models and Kalman Filters (Researchgate); additionally applied PCA, Neural Networks, GMMs, SVMs, K-Means, and Decision Trees for behavioral analysis of eye-tracking data.
- Thesis: *Statistical Methods to Measure Reading Progression Using Eye-Gaze Fixation Points* (link).

Education

M.Sc, Electrical and Computer Engineering — University of Windsor Jan. 2018 – Aug. 2019
Windsor, ON

B.Eng, Mechanical Engineering — University of Windsor Sep. 2012 – Aug. 2016
Windsor, ON